Composite Material Full Test Suite

Interactive MATLAB GUI with virtual tensile, compressive, flexural, impact and vibration tests

MATLAB — Material library — User-defined specimen dimensions — Animated tests

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Overview

`composite_full_test.m` is a single-file MATLAB application for simulating standard mechanical tests on composite materials. The user selects a material from a built-in library, chooses a test type, optionally enters specimen dimensions, and gets a machine-style test report, load–displacement and stress–strain plots, a frequency response, a computed-versus-reported validation list and a live animation of the test. Material properties stay fixed from the library; only specimen geometry changes.

The mechanics behind every test are derived in the accompanying `THEORY.pdf`.

Supported tests

Test	Standard (as labelled)	What the simulation produces
Tensile	ASTM D3039	Stress–strain and load–displacement, UTS, necking and fracture animation
Drop-weight impact	ASTM D7136	Hertzian contact simulation, force vs. displacement, absorbed energy
Charpy impact	ASTM E23	Pendulum animation, impact energy estimate
Izod impact	ASTM D256	Cantilever pendulum animation, impact energy estimate
Flexural (3-point bend)	ASTM D790	Load–deflection, flexural stress–strain
Compressive	ASTM D695	Compressive stress–strain and load–displacement
Vibration	ASTM D3580	Swept-sine response, resonance indicator, time-domain trace

Features

- Start-up dialog for tester name, company/lab and report reference number (editable later)
- Material selector with a status bar summarising UTS, UCS, modulus and natural frequency
- Per-test dimension prompts (width, thickness, gauge/height/span/length as relevant)
- Report listbox and validation listbox, updated when the material or test changes
- Material info pop-up, “compare all” view and tester-info editor
- Dark-themed animation window for each test with live plots

Requirements

- MATLAB with base functionality; the script uses `inputdlg`, `uicontrol`, `uipanel`, `guidata`, `cumtrapz`, `contains` and `yline`, so a recent release (R2018b or later) is recommended
- No toolboxes are called in the code received

Getting started

```
git clone https://github.com/<your-username>/composite-test-suite.git
cd composite-test-suite
```

```
>> composite_full_test
```

How to use

1. Enter tester name, company and reference number in the start-up dialog (cancel to use defaults).
2. Pick a **Material** and a **Test Type** from the drop-downs; the plots and report refresh.
3. Press **GENERATE REPORT** to rebuild the report for the selected test.
4. Press **RUN ANIMATION**, confirm or edit the specimen dimensions, and watch the test.
5. Use **COMPARE ALL**, **MAT. INFO** and **EDIT TESTER** as needed.

Specimen dimension prompts

Test	Inputs requested (defaults from the material library)
Tensile	Width, thickness, gauge length
Compressive	Width, thickness, height (default 1.2× gauge length)
Vibration	Width, thickness, length (default 1.5× gauge length)
Impact (drop-weight, Charpy, Izod)	Width, thickness (length fixed at 80 mm for Charpy/Izod)
Flexural	Width, thickness, support span (default 100 mm)

After entry, area is recomputed as $A = \text{width} \times \text{thickness}$ and the ultimate, yield and break loads as $\sigma A / 10^3$ (kN).

Code structure

```
composite_full_test.m
  composite_full_test()        % main GUI function, nested callbacks
    generate_report_callback   edit_tester        on_sample_change
    show_material_info         get_user_dimensions
    run_animation              % dispatcher
    do_plot_ld / do_plot_ss / do_plot_frequency
    animate_tensile      animate_drop_impact   animate_charpy
    animate_izod         animate_flexural      animate_compressive
    animate_vibration
```

```
% local helpers expected in the full file:
define_material_library, make_stress_strain_curve,
   make_compressive_curve,
build_report_lines, build_validation_lines, style_ax,
make_anim_fig, make_dark_ax, compare_samples
```

Note: the version used to prepare this document ended inside `animate_compressive` and did not contain the helper functions listed above; make sure your repository contains the complete file.

Customising materials

Each library entry is a struct. Fields used by the visible code are:

```
test_no, sample_name, material_class, fibre_matrix
young_modulus (MPa), uts, ucs, yield_strength, break_strength (MPa)
elongation (%), density (kg/m^3), poisson
natural_freq (Hz), damping_ratio
width, thickness, gauge_length (mm), area (mm^2)
ult_load, yield_load, break_load (kN), peak_disp, max_disp (mm)
```

Add an entry in `define_material_library` and it appears in the Material drop-down automatically.

Known issues

1. Pressing Cancel in a dimension dialog raises an error, because defaults are returned as a numeric array but read with `dims{1}`. Return a cell array instead.
2. The flexural span entered by the user is ignored; `animate_flexural` fixes 100 mm. Use the entered value.
3. Drop-weight contact stiffness always hits the 2×10^7 floor, so the result does not vary with material; Charpy and Izod energies are clipped heuristics.
4. Flexural strength is assumed to be 1.5× UTS.
5. Frequency response uses the force-excitation magnification factor and the time trace decay uses the excitation frequency.
6. The header date is hard-coded, and standard designations (E23, D3580) should be verified for composites before citing.

Details and suggested corrections are in `THEORY.pdf`, Section 9.

Roadmap

- Fix the issues above and add input validation (positive dimensions)
- Export reports and plots to PDF/CSV
- Add laminate-level properties (ply orientation, classical lamination theory)
- Replace heuristic impact energies with a pendulum energy balance

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